Towards a Formal Definition of Kinase Inhibitor Selectivity: Empirical Characterization of Definitional Instability and Proposed Desiderata

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Abstract

Kinase inhibitor selectivity is routinely quantified using one of several definitions — including the S-score, selectivity entropy, Gini coefficient, and ratio-based measures — yet these definitions are applied interchangeably without systematic justification. We characterize, across two large kinase inhibitor profiling datasets (Davis, $n = 68$; Klaeger, $n = 222$), the extent and structural sources of disagreement between definitions as their parameters vary. Spearman correlations between median rankings reveal that ratio-based definitions cluster separately from entropy- and distribution-based definitions ($r = 0.27$–0.48 vs. $r = 0.75$–0.91). Rank instability is not random: three mechanistically distinct sources are identified. Drugs with no detected binding above a minimum affinity threshold exhibit maximal instability (mean rank $\sigma \approx 74$) due to definitional noise. Among active compounds, near-tied top targets drive ratio-specific instability ($r = -0.293$, $p < 0.001$), while broad flat binding profiles drive instability within distribution-based families. Attempts to correlate selectivity scores with clinical adverse event rates from FAERS ($n = 46$) and FDA label discontinuation rates ($n = 33$) yield null results, which we attribute to confounding by indication severity and prescription volume. From these findings we derive four desiderata toward a well-formed selectivity measure: a reliability threshold condition, bounded gap sensitivity, distributional consistency under parameter perturbation, and monotonicity under weak off-target addition. These desiderata are intended as the empirical foundation for a future formal axiomatization, in the tradition of Shannon's characterization of entropy.

Introduction

Related Work

Existing Selectivity Metrics and Their Comparison

Several metrics have been proposed to quantify kinase inhibitor selectivity from profiling panel data. Karaman et al.[1] introduced the selectivity score (S-score), defined as the fraction of panel kinases inhibited above a fixed concentration threshold. Uitdehaag and Zaman[2] proposed the selectivity entropy, which treats the normalized binding profile as a probability distribution and computes its Shannon entropy; low entropy indicates activity concentrated on few targets. Graczyk[3] defined a thermodynamics-based partition index estimating the fraction of total binding directed to a reference kinase, suited primarily to hit-to-lead optimization against a specified target. Klaeger et al.[4] introduced the concentration- and target-dependent selectivity (CATDS) score, which quantifies the reduction in binding to a given target relative to the summed reduction across all detected targets at a specified inhibitor concentration. Bosc et al.[5] subsequently proposed two further metrics — the window score and ranking score — and performed a comparative analysis of all five definitions on the Davis and Anastassiadis datasets, showing that compound rankings differ substantially depending on the metric applied and that certain compounds cannot be scored under some definitions when threshold conditions are not satisfied.

Reconciliation attempts have remained limited in scope. Bajorath et al.[6] compared CATDS-based selectivity assessments from Klaeger et al. with activity data mined from ChEMBL for the same compounds, finding broad consistency between the two sources but noting that no simple relationship between selectivity and clinical performance was apparent. The KInhibition platform[7] explicitly acknowledged that multiple metrics yield conflicting results for the same compound but addressed this through a unified query interface rather than theoretical analysis.

A common limitation of these studies is that each metric is evaluated at fixed parameter values. The sensitivity of rankings to parameter choice — the activity threshold in the

S-score, the baseline in the entropy and Gini coefficient, the reference off-target in ratio-based measures — has not been systematically characterized. The structural properties of binding profiles that determine when definitions agree or disagree have not been identified. The present work addresses both gaps.

Axiomatic Approaches to Ranking and Measurement

Axiomatic characterization of measurement functions is well-established in adjacent fields. Shannon[8] demonstrated that the entropy of a probability distribution is uniquely determined, up to a multiplicative constant, by four requirements: continuity, maximality on the uniform distribution, expansibility, and additivity over independent subsystems. Khinchin[9] subsequently strengthened this uniqueness result. The selectivity entropy of Uitdehaag and Zaman applies the Shannon formula to normalized binding profiles but provides no analogous axiomatic justification; it is not known whether it is the unique function satisfying reasonable selectivity-specific desiderata, or one of many equivalent alternatives.

Arrow[10] established that no social welfare function satisfies simultaneously the requirements of unrestricted domain, Pareto efficiency, independence of irrelevant alternatives, and non-dictatorship. The theorem is relevant here in two respects: it demonstrates that axiomatic analysis of a ranking concept can yield impossibility rather than uniqueness, and it shows that competing ranking functions reflect genuine tradeoffs between properties that cannot all be achieved simultaneously. Pavillidis et al.[11] applied this methodology to truth discovery — the problem of identifying true facts from conflicting reports of unknown reliability — posing it as a social choice problem, formulating candidate axioms, and proving that a subset cannot be jointly satisfied. The structure of that problem is closely analogous to selectivity measurement, where multiple definitions produce conflicting rankings and the question of which to trust is unresolved.

Formal methods have been applied to biological systems primarily in the context of process dynamics. Cardelli[12] used the pi-calculus to model molecular biology, and the

Beacon Calculus[13] addresses stochastic reaction networks; both verify properties of temporal execution rather than static measurements. Type-theoretic approaches to scientific units of measure, including Kennedy's[14] F# units-of-measure system and ongoing formalization in Lean 4 Mathlib,[15] enforce dimensional consistency for physical quantities but do not extend to biochemical properties. To our knowledge, axiomatic analysis has not previously been applied to any selectivity or promiscuity measure in drug discovery. The present work proposes empirically motivated desiderata as a foundation for future formal axiomatization.

Methods

Datasets

Two publicly available kinase inhibitor profiling datasets were used. The Davis dataset[16] comprises binding affinities (K_d) for 68 small-molecule kinase inhibitors measured against 433 human kinase domains. Affinity values are reported as dissociation constants and were converted to $\mathrm{p}K_d = -\log_{10}(K_d/10^9)$ prior to analysis, following the convention of the DeepDTA benchmark.[17] Pairs for which no binding was detected at the screening concentration of 10 μM were assigned $\mathrm{p}K_d = 5.0$, corresponding to a K_d of 1 mM, which represents the detection floor of the assay. The resulting affinity matrix is complete (68×433, 29,444 entries).

The Klaeger dataset[4] comprises binding affinities for 243 clinical kinase inhibitors profiled against the human kinome using Kinobeads chemical proteomics. Apparent dissociation constants (K_d^{app}) were derived from dose–response curves across eight inhibitor concentrations in cancer cell lysates. Only high-confidence measurements (as annotated by the original authors) were retained, yielding 5,123 drug–target pairs for 222 drugs and 343 kinase targets. Affinity values were converted to $\mathrm{p}K_d$ using the same formula as above. Missing entries, representing kinase–inhibitor combinations not detected in the assay, were assigned the background value $\mathrm{p}K_d = 5.0$. The resulting matrix (222×343) has 17.9% active entries (pKd > 6, i.e. $K_d < 1$ μM) for Davis and 3.6% for Klaeger, reflecting the more stringent detection threshold of the chemoproteomic assay.

Selectivity Definitions and Parameterization

Four selectivity definitions were implemented, each controlled by one or more parameters whose values are not standardized in the literature.

The **S-score**[1] is defined as

$$S(\tau) = \frac{1}{N} \sum_{i=1}^{N} \mathbf{1}[\mathrm{p}K_{d,i} > \tau], \tag{1}$$

where N is the number of kinases in the panel and τ is an activity threshold. Lower values indicate greater selectivity. We swept $\tau \in \{5.5, 5.75, \ldots, 8.0\}$ (11 values), spanning K_d cutoffs from approximately 300 nM to 10 nM.

The **selectivity entropy**[2] treats the baseline-subtracted binding profile as a probability distribution:

$$H(\beta) = -\sum_{i=1}^{N} p_i \log_2 p_i, \quad p_i = \frac{\max(\mathrm{p}K_{d,i} - \beta,\, 0)}{\sum_j \max(\mathrm{p}K_{d,j} - \beta,\, 0)}, \tag{2}$$

where β is a baseline affinity below which binding is considered negligible. Lower entropy indicates greater selectivity. We swept $\beta \in \{5.0, 5.25, \ldots, 6.5\}$ (7 values).

The **Gini coefficient**[3] of the baseline-subtracted profile is

$$G(\beta) = \frac{2 \sum_{i=1}^{N} i \cdot x_{(i)}}{N \sum_{i=1}^{N} x_{(i)}} - \frac{N+1}{N}, \tag{3}$$

where $x_{(i)} = \max(\mathrm{p}K_{d,i} - \beta,\, 0)$ sorted in ascending order. Higher values indicate greater selectivity. We swept β over the same range as for the entropy.

The **ratio** definition measures the gap between the primary target and a specified off-target:

$$R(k) = \mathrm{p}K_{d,(1)} - \mathrm{p}K_{d,(k+1)}, \tag{4}$$

where $\mathrm{p}K_{d,(j)}$ denotes the j-th largest affinity in the profile and $k \in \{1, 2, 3, 4, 5\}$ indexes the off-target used as reference. Higher values indicate greater selectivity.

Rank Stability Analysis

For each definition and each parameter value, compounds were ranked from most selective (rank 1) to least selective (rank n, where n is the number of compounds in the dataset). Ties

were broken by average rank. This yielded 30 rank vectors per dataset: 11 for the S-score, 7 for entropy, 7 for Gini, and 5 for ratio.

For each compound, rank instability was quantified as the standard deviation of its rank across all 30 configurations (overall instability) and separately across configurations within each definition family (within-family instability). Cross-definition agreement was assessed using Spearman rank correlation between the median rank vectors of each definition family. The median was taken across parameter values within each family prior to correlation to obtain a single representative ranking per definition.

Associations between rank instability and binding profile features were assessed using Spearman correlation. Profile features computed for each compound included: the number of active kinases ($\mathrm{p}K_d > 6$), the gap between the top and second-ranked affinities, the standard deviation and range of active affinities, and the absolute affinity of the second-ranked target. All correlations are reported with two-tailed p-values; significance thresholds of $p < 0.05$, $p < 0.01$, and $p < 0.001$ are denoted *, **, and *** respectively. Analyses were performed in Python 3.10 using NumPy, pandas, and SciPy.[18–20]

Clinical Outcome Data

For the subset of Klaeger compounds that are FDA-approved ($n = 46$ after excluding drugs with no FAERS records), adverse event report counts were retrieved from the FDA Adverse Event Reporting System (FAERS) via the openFDA API. Total report counts, serious event counts, and death and hospitalization subcounts were retrieved per drug. Serious event rates and death rates were computed by normalizing subcounts by total reports per drug.

Treatment discontinuation rates due to adverse events were extracted from FDA drug labels retrieved via the openFDA label API for 33 approved drugs with available adverse reactions sections. Label text was reviewed manually to identify the primary clinical trial discontinuation rate attributable to adverse reactions.

Associations between selectivity scores and each clinical outcome measure were assessed

using Spearman correlation across drugs with available data.

Results

Selectivity Definitions Cluster into Two Families

Across 30 parameter configurations (11 S-score thresholds, 7 entropy baselines, 7 Gini baselines, 5 ratio off-target indices), compound rankings were computed for all drugs in both datasets. Spearman correlations between the median ranking of each definition family reveal a consistent two-cluster structure (Table 1). On the Davis dataset, S-score, entropy, and Gini are mutually correlated ($r = 0.91$–0.999), while ratio correlates substantially less with each ($r = 0.34$–0.45). On the larger Klaeger dataset, the same structure is preserved but with lower within-cluster correlations ($r = 0.75$–0.91), reflecting greater diversity of binding profiles among the 222 clinical compounds. The ratio definition is the consistent outlier in both datasets, with cross-family correlations of $r = 0.27$–0.62.

Table 1: Spearman correlations between median rankings by definition family. Upper triangle: Davis ($n = 68$). Lower triangle: Klaeger ($n = 222$).

	S-score	Entropy	Gini	Ratio
S-score	—	0.949	0.954	0.454
Entropy	0.910	—	0.999	0.343
Gini	0.745	0.869	—	0.343
Ratio	0.265	0.480	0.617	—

This two-family structure reflects a fundamental difference in what the definitions measure. The S-score, entropy, and Gini all quantify the concentration of activity across the full kinase panel; they differ in functional form but respond to the same underlying property of the binding profile. The ratio definition, by contrast, quantifies the gap between the primary target and a specific off-target, ignoring the remainder of the binding profile entirely. A compound that strongly inhibits one kinase and moderately inhibits twenty others will be ranked as highly selective by ratio (large gap to the second target) but as moderately promiscuous by entropy and Gini (activity distributed across many targets).

Rank Instability Is Structured and Predictable

Overall rank instability — measured as the standard deviation of a compound's rank across all 30 configurations — varies substantially across compounds (Klaeger: mean $\sigma = 35.4$, range 7.1–91.6; Davis: mean $\sigma = 8.5$, range 1.6–19.3). Instability is not uniformly distributed: within each definition family, specific binding profile features predict which compounds receive unstable rankings (Table 2).

Within the ratio family, the gap between the top and second-ranked affinities (top_1–top_2 gap) is the strongest predictor of instability ($r = -0.446$, $p < 0.001$ on Davis; $r = -0.293$, $p < 0.001$ on Klaeger). Compounds with near-identical affinities for their two best-bound kinases receive highly variable ratio rankings because the choice of reference off-target ($k = 1$ vs. $k = 2, \ldots, 5$) determines whether the compound appears selective or promiscuous.

Within the entropy and Gini families, the number of active kinases ($\text{p}K_d > 6$) negatively predicts instability ($r = -0.279$, $p < 0.05$ for entropy on Davis; $r = -0.474$, $p < 0.001$ on Klaeger). Compounds with many active kinases have well-defined broad profiles whose entropy and Gini scores are stable across baseline choices; compounds with few active kinases are sensitive to the baseline because small changes can move targets in or out of the active set, altering the distribution substantially. The active affinity range is also a significant predictor across all three distribution-based definitions on Klaeger ($r = -0.31$ to -0.37, all $p < 0.001$), consistent with the interpretation that heterogeneous profiles are easier to rank stably than flat ones.

S-score instability is not significantly predicted by any profile feature on either dataset, suggesting that it is driven primarily by where compounds fall relative to the threshold rather than by intrinsic profile shape.

Three Mechanistically Distinct Sources of Instability

Examination of the most and least unstable compounds in the Klaeger dataset reveals three structurally distinct sources of rank instability, each implicating a different aspect of how the

Table 2: Spearman correlations between binding profile features and within-family rank instability on the Klaeger dataset ($n = 206$ active drugs). Significance: * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$.

Feature	S-score	Entropy	Gini	Ratio
n_{active}	-0.263^{***}	-0.474^{***}	-0.332^{***}	-0.091
top_1–top_2 gap	$+0.044$	$+0.108$	$+0.068$	-0.293^{***}
Active affinity std	-0.302^{***}	-0.263^{***}	-0.319^{***}	-0.131
Active affinity range	-0.366^{***}	-0.350^{***}	-0.309^{***}	-0.141^{*}
top_2 pK_d	-0.312^{***}	-0.309^{***}	-0.390^{***}	$+0.112$

definitions are constructed.

Type 1 — Zero-active compounds. The five most unstable Klaeger compounds (e.g. AZD-8055, BMS-911543, Filgotinib; rank $\sigma > 80$) all have zero active kinases at the p$K_d > 6$ threshold. These compounds have no meaningful binding to any profiled kinase at sub-micromolar affinity, yet all four definitions still assign them numerical selectivity scores based on the detection floor values. The resulting rankings are arbitrary: different parameterizations produce different orderings of compounds that are, in practice, indistinguishable from one another. Mean rank instability for zero-active compounds is $\sigma \approx 74$, compared with $\sigma \approx 32$ for compounds with at least one active kinase ($p < 0.001$, Mann–Whitney U test). This pattern is observed only in the Klaeger dataset; all Davis compounds have at least one active kinase, consistent with Davis's fixed-concentration screening design, which reports a K_d only when activity is first detected at 10 μM.

Type 2 — Near-tied top targets. Among active compounds, ratio-specific instability is strongly predicted by a small top_1–top_2 gap ($r = -0.293$, $p < 0.001$). Compounds such as Dovitinib (top_1–top_2 gap = 1.935 pKd units) and BMS-690514 (gap = 1.985) show large entropy-ratio disagreement despite having well-defined primary targets, because the ratio definition is sensitive to whether the reference off-target is the second or a lower-ranked kinase. By contrast, Drug 23 in the Davis dataset (top_1–top_2 gap = 0.019) shows maximal ratio instability because its two best-bound kinases are essentially indistinguishable, making the ratio highly sensitive to which is designated the primary target.

Type 3 — Broad profiles with a dominant primary target. The largest entropy-ratio disagreements in the Klaeger dataset arise for compounds with high n_{active} and a large top_1–top_2 gap (e.g. Dovitinib: $n_{\text{active}} = 26$, gap = 1.935; BMS-690514: $n_{\text{active}} = 20$, gap = 1.985). These compounds have one clearly dominant target but also bind many kinases at moderate affinity. The ratio definition scores them as highly selective (large gap between primary and secondary target), while the entropy and Gini definitions score them as moderately promiscuous (activity spread across many targets). Neither characterization is incorrect; they reflect genuinely different questions about selectivity.

Conversely, the most stable compounds in both datasets (e.g. XL-228, $\sigma = 7.1$; PF-03814735, $\sigma = 8.1$) are highly promiscuous, with 40–57 active kinases. When a compound binds nearly the entire profiled kinome, all definitions rank it as non-selective regardless of parameterization, producing low instability despite — or rather because of — high promiscuity.

Clinical Outcome Correlations

Spearman correlations between median selectivity rankings and FAERS-derived adverse event rates (serious event rate, death rate, hospitalization rate) across 46 FDA-approved Klaeger compounds were uniformly weak and non-significant (all $|r| < 0.18$, all $p > 0.14$). Similarly, correlations between selectivity rankings and FDA label discontinuation rates due to adverse events ($n = 33$ drugs) were non-significant across all four definition families (all $|r| < 0.27$, all $p > 0.13$).

We attribute these null results to confounding inherent in both data sources rather than absence of a biological relationship between selectivity and toxicity. FAERS report counts are driven primarily by prescription volume and indication prevalence rather than drug toxicity per se; drugs approved for common conditions (e.g. tofacitinib for rheumatoid arthritis, palbociclib for breast cancer) accumulate large report counts regardless of their adverse event profiles relative to alternative therapies. Discontinuation rates from drug labels reflect the clinical trial populations and indication-specific toxicity expectations for those populations,

making cross-indication comparisons unreliable. A rigorous test of the relationship between in vitro selectivity and clinical toxicity would require indication-stratified, treatment-related adverse event rates from randomized controlled trials with matched comparators, which are not available in a standardized form across the breadth of kinase inhibitors studied here.

Desiderata for a Well-Formed Selectivity Measure

The empirical findings of Section motivate four properties that any well-formed kinase inhibitor selectivity measure should satisfy. We present these as desiderata rather than axioms: they are necessary conditions motivated by observed failure modes of existing definitions, but they are not yet proven to be jointly sufficient to characterize a unique measure, nor is it known whether they are jointly satisfiable. Establishing these properties formally is the subject of future work.

Let $\mathbf{x} = (x_1, \ldots, x_N) \in \mathbb{R}^N$ denote a compound's binding profile across N kinases, where $x_i = \mathrm{p}K_{d,i}$. A selectivity measure is a function $f : \mathbb{R}^N \to \mathbb{R}$, where higher values indicate greater selectivity. A selectivity ranking over a set of compounds is induced by ordering their f-scores.

D1 — Reliability threshold. There exists a threshold $\tau^* > 0$ such that $f(\mathbf{x})$ is undefined, or explicitly flagged as unreliable, when $\max_i x_i \leq \tau^*$.

Motivation. Compounds with no detected binding above the assay detection floor receive arbitrary selectivity scores under all four definitions studied here, producing rank instability with mean $\sigma \approx 74$ compared with $\sigma \approx 32$ for active compounds (Section , Type 1). Selectivity is a property of binding; a compound that does not bind cannot be meaningfully ranked relative to compounds that do. No existing definition incorporates an explicit reliability condition: both the S-score and selectivity entropy assign scores to compounds with no sub-micromolar binding, treating detection floor values as informative.

D2 — Bounded gap sensitivity. For any compound profile $\mathbf{x}$ satisfying D1, a small perturbation to the identity of the reference off-target should produce a bounded change in $f(\mathbf{x})$. Formally, if f depends on the k-th ranked affinity $x_{(k)}$, then $|f(\mathbf{x}; k) - f(\mathbf{x}; k+1)|$ should be bounded by a function of $x_{(k)} - x_{(k+1)}$.

Motivation. Ratio-based definitions violate this property when $x_{(1)} \approx x_{(2)}$: a compound

with near-identical affinities for its top two kinases receives a large ratio score when $k = 1$ (the gap to the next off-target may be large) but a near-zero score when $k = 2$ (the gap between the tied targets is negligible). This produces the Type 2 instability identified in Section , where top_1–top_2 gap is the strongest predictor of ratio instability ($r = -0.293$, $p < 0.001$). Distribution-based definitions (entropy, Gini) satisfy a weaker version of this property because they do not privilege any single pairwise comparison.

D3 — Distributional consistency. The ranking induced by f should be stable under small perturbations to the activity baseline β used to define the active binding distribution. Formally, for any two compounds $\mathbf{x}, \mathbf{y}$ and any $\epsilon > 0$, there exists $\delta > 0$ such that if $|\beta_1 - \beta_2| < \delta$ then the ranking of $\mathbf{x}$ relative to $\mathbf{y}$ under $f(\cdot; \beta_1)$ agrees with that under $f(\cdot; \beta_2)$.

Motivation. The entropy and Gini definitions are parameterized by a baseline β that determines which affinities contribute to the binding distribution. Compounds with few active kinases are sensitive to this choice because small changes in β can move individual kinases in or out of the active set, substantially altering the distribution. The negative correlation between n_{active} and entropy/Gini instability ($r = -0.47$ on Klaeger, $p < 0.001$) reflects this sensitivity. D3 requires that the ranking be robust to the choice of β within a reasonable neighborhood, which existing implementations do not guarantee.

D4 — Monotonicity under weak off-target addition. If compound $\mathbf{x}$ has a weak additional binding interaction added — formally, if $\mathbf{x}' = (x_1, \ldots, x_N, x_{N+1})$ where $x_{N+1} \leq \tau^*$ — then $f(\mathbf{x}') \leq f(\mathbf{x})$. Adding a sub-threshold binding interaction should not increase a compound's selectivity score.

Motivation. Ratio-based definitions can violate this property: if the new weak interaction becomes the reference off-target ($k = 1$), the ratio to the primary target may increase, artifactually improving the apparent selectivity score. This failure mode is practically relevant when profiling panels differ in size or composition across experiments, as is common when comparing selectivity across different assay platforms or across datasets such as Davis and

Klaeger. Distribution-based definitions satisfy D4 naturally, since adding a sub-threshold term to the probability distribution decreases or preserves entropy and Gini.

Table 3 summarizes which existing definitions satisfy each desideratum. No existing definition satisfies all four simultaneously. Whether a definition satisfying all four exists, and whether the four desiderata are jointly consistent, are open questions that we identify as the primary targets of a formal axiomatization.

Table 3: Satisfaction of desiderata by existing selectivity definitions. ✓: satisfied by construction. ∼: partially satisfied or satisfied under additional assumptions. ×: violated in general.

Desideratum	S-score	Entropy	Gini	Ratio
D1 (reliability threshold)	×	×	×	×
D2 (bounded gap sensitivity)	∼	✓	✓	×
D3 (distributional consistency)	∼	×	×	✓
D4 (monotonicity, weak off-target)	✓	✓	✓	×

Discussion

Conclusion

Acknowledgement

The authors thank the developers of the Davis and Klaeger datasets for making their data publicly available.

Supporting Information Available

Code and data are available at `https://github.com/polinavino/kinase-selectivity-definitions`.

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